



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 1 Report

Internship Program — Batch 2026

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1. Introduction

This report summarizes the activities completed for Experiment 09: Deep Learning – Neural Networks & Backpropagation under the M-Tech Production & Marketing AI/ML Internship Program 2026. The experiment introduced the principles of Deep Learning and explained how Artificial Neural Networks imitate the human brain to recognize patterns and solve complex problems. It also focused on the learning mechanism known as backpropagation, which helps neural networks improve their predictions over time.

2. Objectives

- Gain an understanding of Deep Learning concepts.
 - Study the architecture of Artificial Neural Networks.
 - Learn the purpose of neurons, weights, and hidden layers.
 - Understand how backpropagation trains neural network models.
 - Explore practical applications of neural networks in AI.
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3. Task Description

This experiment involved studying the internal structure of Artificial Neural Networks and the complete learning cycle of a neural network model. The experiment explained how input data passes through different layers, how prediction errors are calculated, and how the model adjusts its parameters using backpropagation to improve accuracy.

4. Work Done

During this experiment, I explored the basic components of neural networks, including neurons, weights, biases, and activation functions. I learned how information travels through multiple network layers during forward propagation and how prediction errors are minimized through backpropagation. I also studied the importance of activation functions such as ReLU, Sigmoid, and Softmax in enabling neural networks to solve non-linear problems. Through practical examples, I understood how repeated training helps improve the overall performance of deep learning models.

5. Challenges Faced

- Understanding how gradients are calculated during backpropagation.
 - Learning the mathematical concepts involved in weight optimization.
 - Distinguishing between different activation functions.
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6. Key Learnings

- Artificial Neural Networks are the foundation of Deep Learning.
- Backpropagation continuously improves model accuracy.
- Hidden layers help neural networks learn complex relationships.
- Activation functions allow models to solve non-linear problems.
- Deep Learning has applications in computer vision, NLP, healthcare, and robotics.

7. Conclusion

This experiment strengthened my understanding of neural network architecture and the training process used in Deep Learning. I developed a clear understanding of how neural networks learn from data and why they are widely used in modern Artificial Intelligence systems

Intern Signature:



Supervisor:

Name:

Ghazi Muhammad Abdullah

Name:

Date:

13 July 2026

Designation: